



MicroHack: Microsoft Fabric Lakehouse End-to-End

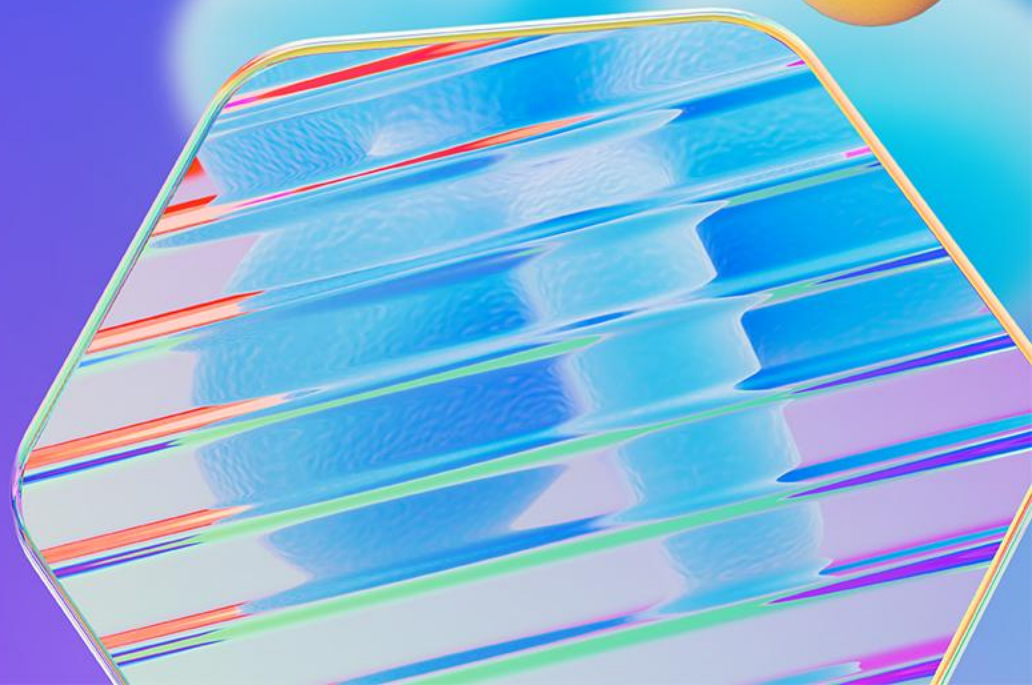
Erik Eckhardt | Filip Slánička



Agenda

- Introduction to the MicroHack
- Introduction to the Microsoft Fabric Solution
- MicroHack: Microsoft Fabric Lakehouse End 2 End
- Let's Start Hacking

Introducing to the MicroHack

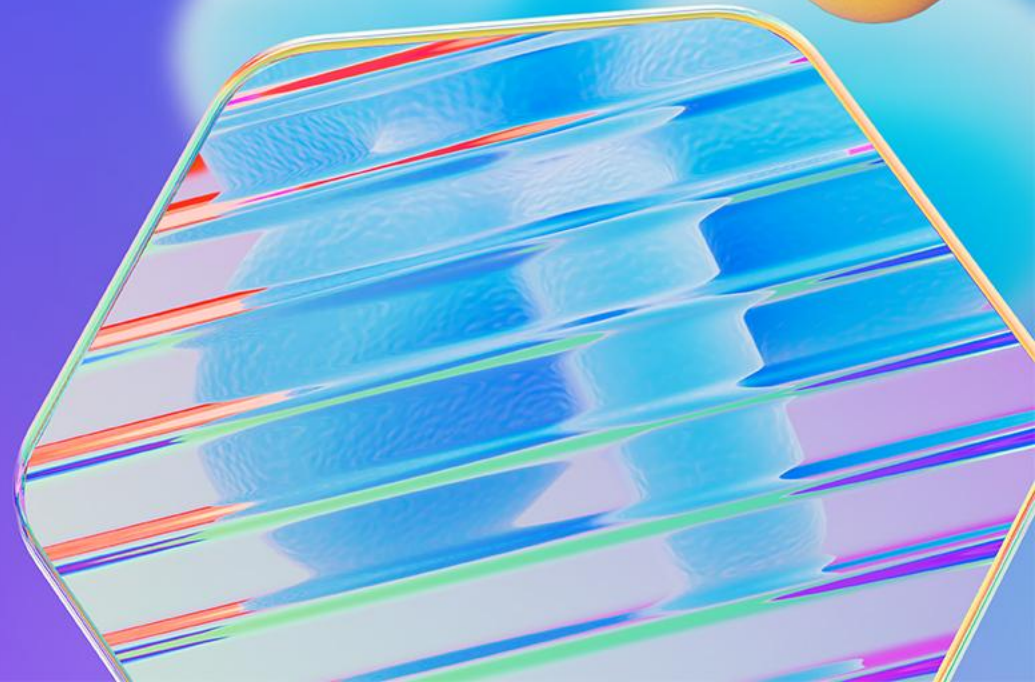


Introducing to the MicroHack

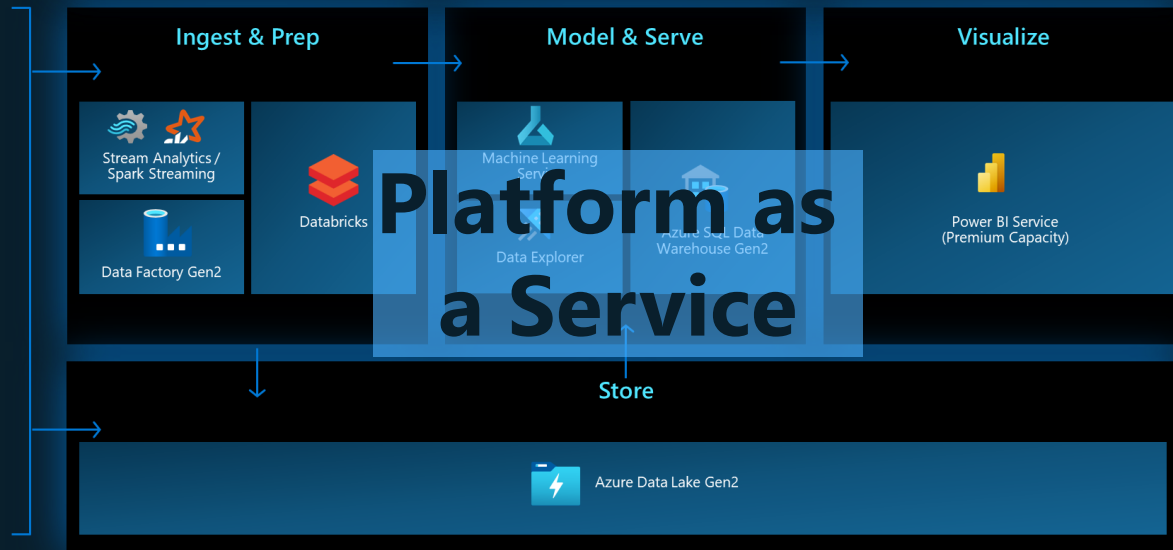
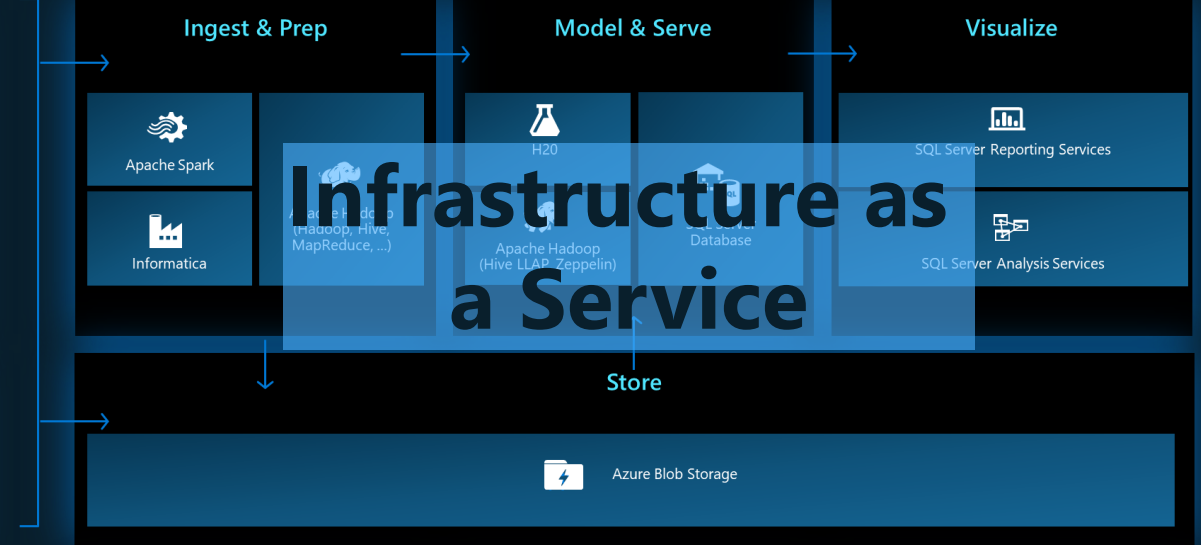
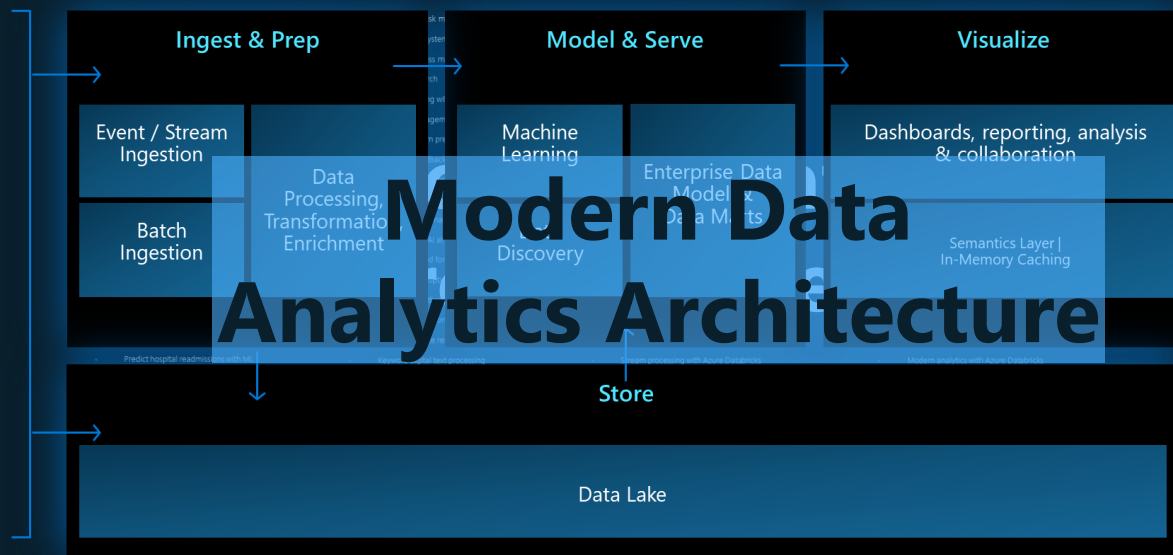
Concept

- Challenge-Based Tutorial Format
 - You'll be actively involved in coding, implementation, and problem-solving.
- What to Expect
 - Real-world scenarios and exercises with direct application of concepts in live environments.
- Hands-On Approach
 - Please read the detailed descriptions provided for each challenge.
 - Carefully review the goals, required actions, and success criteria.
- Support and Collaboration
 - Instructors are here to guide you. Feel free to ask questions and support your fellow participants.
 - A complete end-to-end solution guide with screenshots is also available.
- Outcome
 - By the end of this MicroHack, you'll gain practical experience and new skills that you can directly apply to your projects.

Introducing the Microsoft Fabric solution



Microsoft Data Analytics Evolution





Microsoft Power BI

+



Azure Synapse Analytics



BI App



Live dashboards



Visualizations



Reports

01001
10101

Datasets

Shared Workspaces

PRO | Premium Per User | Premium Capacity

Cloud-based BI services

+



Azure Synapse Analytics

Studio

Data Integration

Management

Monitoring

Security

Analytics runtimes



SQL



Apache Spark



Data Explorer



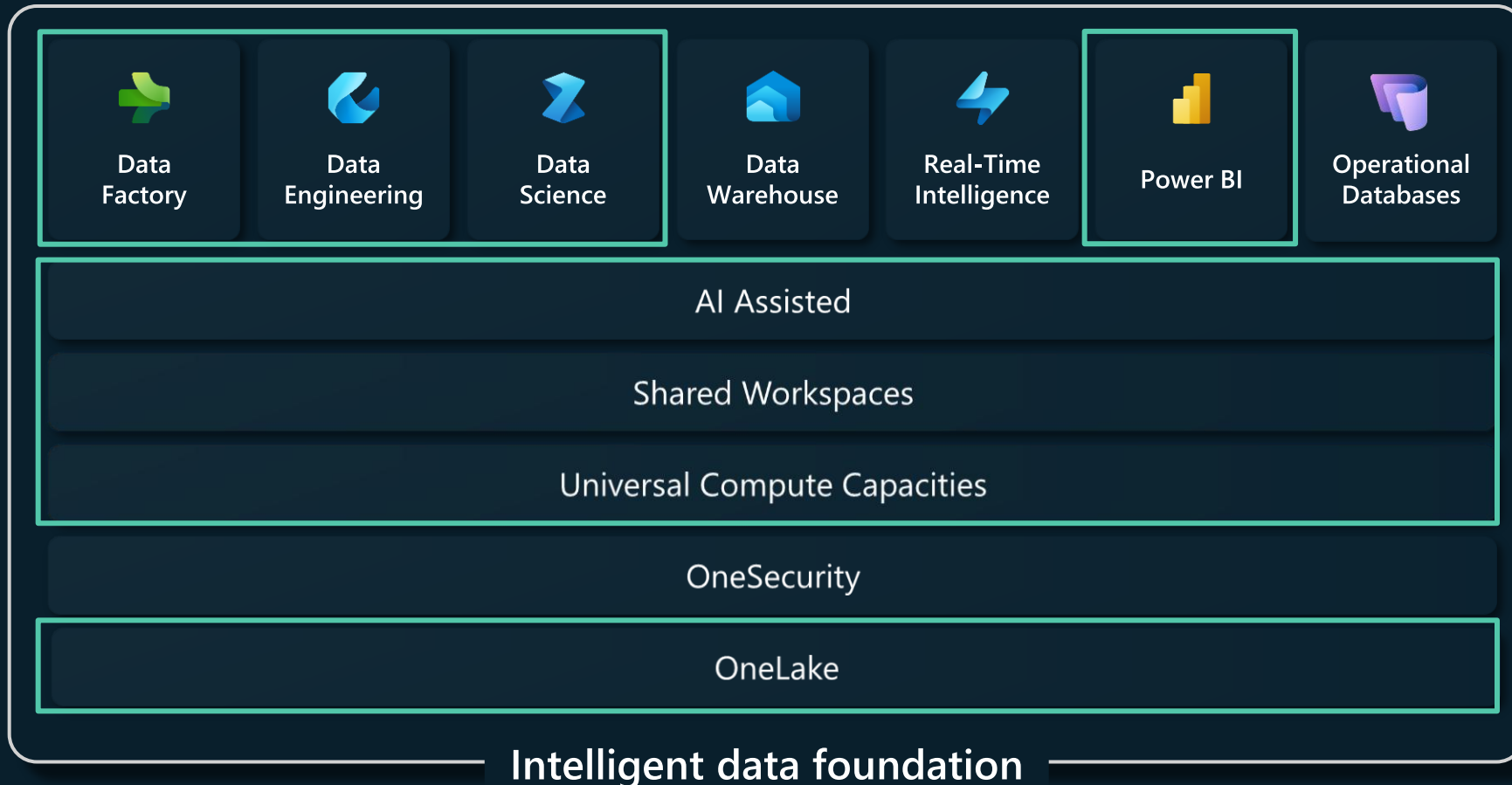
Azure Data Lake Storage Gen2

Cloud-based
Analytics services



Microsoft Fabric

The unified data platform for the era of AI



SaaS with single...

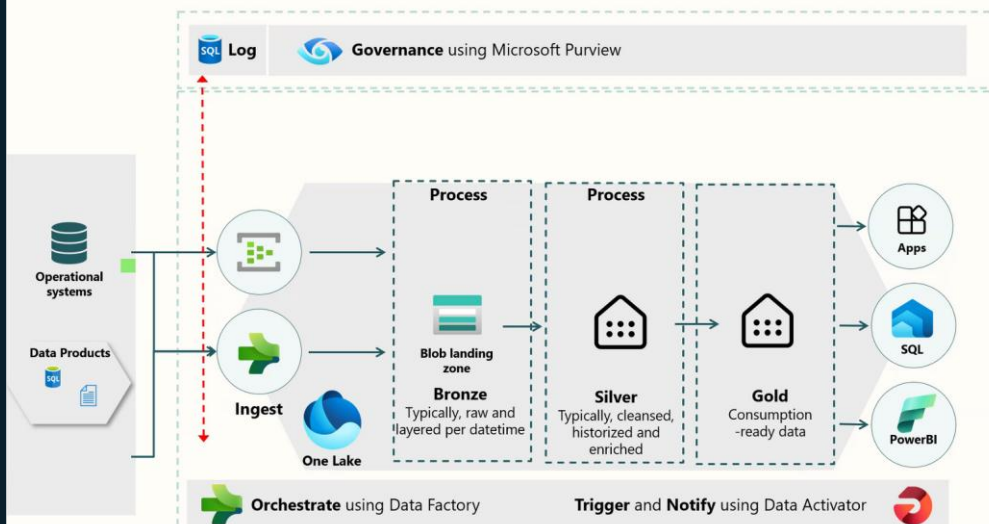
- Onboarding and trials
- Sign-on
- Navigation model
- UX model
- Workspace organization
- Collaboration experience
- Data Lake
- Storage format
- Data copy for all engines
- Security model
- CI/CD
- Monitoring hub
- OneLake Catalog
- Governance & compliance
- Domains



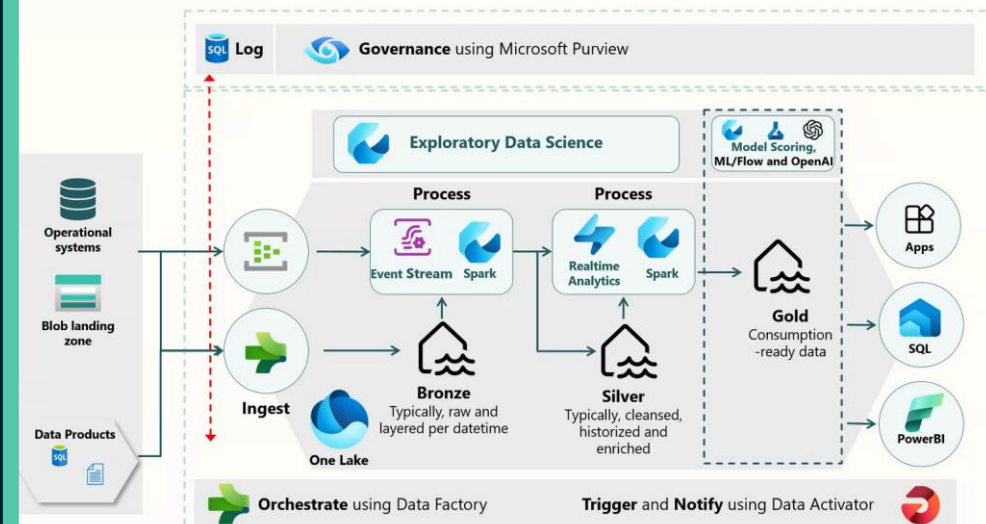
Microsoft Fabric

Data Architecture Patterns

The Warehouse Pattern



The Lakehouse Pattern





Microsoft Fabric

A lakehouse and a medallion architecture

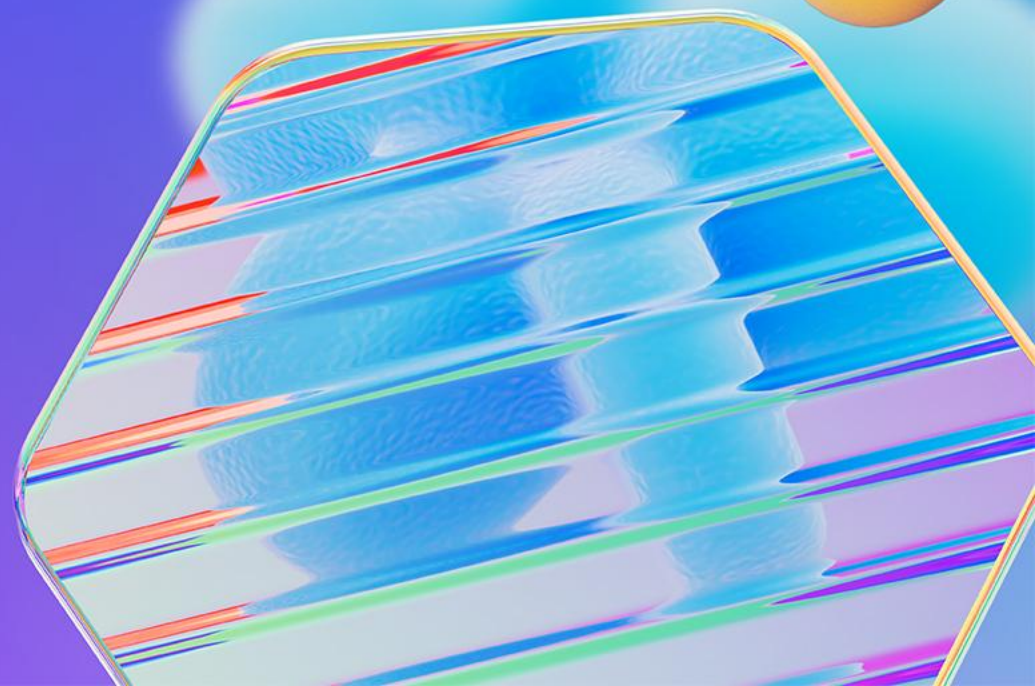


- A lakehouse is an open architecture that combines the best elements of data lakes and data warehouses.
- Lakehouses are enabled by a new system design: implementing similar data structures and data management features to those in a data warehouse directly on top of low-cost cloud storage in open formats with decoupled computes.



- A medallion architecture is a data design pattern used to logically organize data in a lakehouse.
- The goal is to incrementally and progressively improve the structure and quality of data as it flows through each layer of the architecture.
- From a data modeling perspective the Silver Layer has more 3NF write-performant data models, the Gold Layer is more de-normalized and read-optimized data models Kimball style star schema-based or Inmon style Data marts.

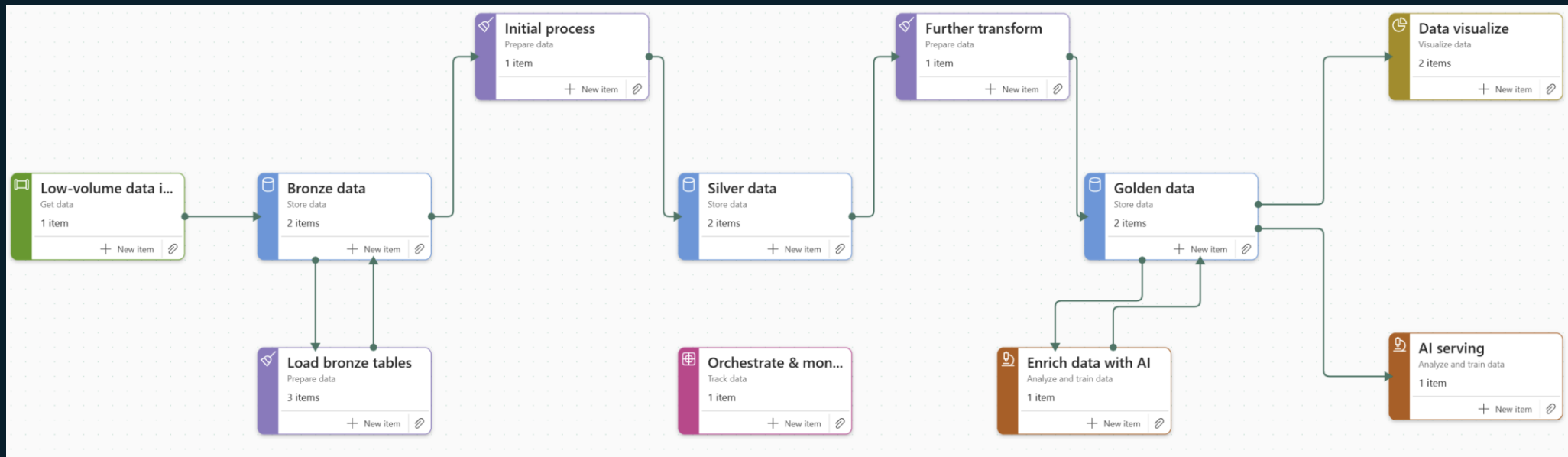
MicroHack: Microsoft Fabric Lakehouse End-to-End



MicroHack: Microsoft Fabric Lakehouse End-to-End

Context & objectives

- Context
 - Learn how to integrate, transform, and manage data in a lakehouse using Fabric workloads such as Data Factory, Data Engineering, Data Science, and Power BI.
- Objectives
 - Develop a complete data workflow: ingestion, preparation, transformation and serving.



MicroHack: Microsoft Fabric Lakehouse End-to-End Challenges

📌 Important

ch01: Prepare the infrastructure

[Challenge](#) | [Solution](#)

ch02: Ingest *raw* data

[Challenge](#) | [Solution](#)

ch03: Load *bronze* tables

[Challenge](#) | [Solution](#)

ch04: Transform data using notebooks and Spark clusters

[Challenge](#) | [Solution](#)

ch05: Build *gold* tables and a semantic layer for Power BI reporting

[Challenge](#) | [Solution](#)

ch06: Create a master pipeline for orchestration

[Challenge](#) | [Solution](#)

ch07: Process unstructured data with an LLM

[Challenge](#) | [Solution](#)

ch08: Data Agent: Talk with your data

[Challenge](#) | [Solution](#)

Extra exercises

[Solution](#)

- Focus on implementing **at least the first 6 challenges** today as a minimum out of the 8 major challenges.

MicroHack: Microsoft Fabric Lakehouse End-to-End

Pre-req. for challenges

⚠ Caution

Before starting the challenges, make sure to go through the step-by-step guide for [New Azure User Setup](#).

- You will receive “**XX**” now for your credentials. Please remember it!

Credentials

Login	Password
UserXX@fabric.microhack.org	dummyHeslo1234

MicroHack: Microsoft Fabric Lakehouse End-to-End

Ch01: Prepare the infrastructure for Microsoft Fabric

Goal

In this challenge, you will prepare the Microsoft Fabric environment for the upcoming challenges. Each participant should get their own Microsoft account for Azure and Office 365. Using your account, create a Fabric capacity and workspace, attach the capacity to your workspace, and create the first lakehouse for the [bronze layer](#) of the [medallion architecture](#).

Actions

- Deploy MS Fabric Capacity into pre-defined Resource Group "rg-MicroHackFabricCapacities" and any available region
 - Name: **yoursurname0fabric** with size **F4**
 - e.g. novak0fabric
- Create Workspace in MS Fabric: **YourSurname NY Taxi Analysis [DEV]**
 - e.g. Novak NY Taxi Analysis [DEV]
- Add your Fabric Capacity to your Workspace.
- Change Spark settings: **Use 1 node only**
- Create the 1st lakehouse: **bronzerawdata**

Success Criteria

- MS Fabric Capacity is running.
- Workspace is ready to solve your workloads.
- The 1st Lakehouse and SQL analytics endpoint is deployed and you can connect

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

NY Taxi Analysis [DEV]

+ New item

New folder

Import

Migrate

Ch Select

Name	Type	Task
bronzerawdata	Lakehouse	—
bronzerawdata	SQL analytics endpoint	—

Microsoft Fabric

Contoso

What platform metrics ap

Create

Manage view

Refresh

Export to CSV

You are viewing a new version of Browse experience. Click here to acce

Filter for any field...

Subscription equals all

Resou

Name	Type
eec0fabric	Fabric Capacity

- User01 - User06 > Region: West Europe
- User07 - User12 > Region: North Europe
- User13 - User18 > Regions: Sweden Central
- User19 - User25 > Region: West US 3

MicroHack: Microsoft Fabric Lakehouse End-to-End

Ch02: Ingest raw data with data pipelines and shortcuts

Goal

Focus on integrating two data sources from NYC taxi data: one from 2015 and the other from 2023. The table below presents the metaphors we will use to integrate the data:

Actions

- Create Pipeline: Ingest Green Taxi 2015
 - With Copy Activity to Ingest Data:
 - Source: Azure Blob Storage: <https://stmsfabricmicrohacksw.blob.core.windows.net/>
 - Authentication kind: **Organizational account**
 - File path:
 - Container: **taxidata**
 - File: **green_tripdata_2015-01.parquet**
 - File format: **Binary**
 - Destination : Lakehouse **bronzerawdata**
 - File path directory: **Landing/GreenTaxi2015**
 - Format: **Binary**
- Create Shortcut in your Lakehouse into **Files > Landing** to **2023** Folder in Azure Data Lake Gen2:
 - <https://adlstmsfabricmicrohacksw.dfs.core.windows.net/>
 - Authentication kind: **Organizational account**

Success Criteria

- You have File **"green_tripdata_2015-01.parquet"** in your Lakehouse: **Files > Landing > GreenTaxi2015**
 - You can see the content of the file
- You have shortcut to Folder **"2023"** in your Lakehouse with files **"NYC-Taxi-Discounts-Per-Day.csv"** and **"green_tripdata_2023-01.parquet"**
 - You can see the content of both files

Home

Get data ▾ New semantic model Open notebook ▾ Manage OneLake data access (preview) Update all variables

Explorer

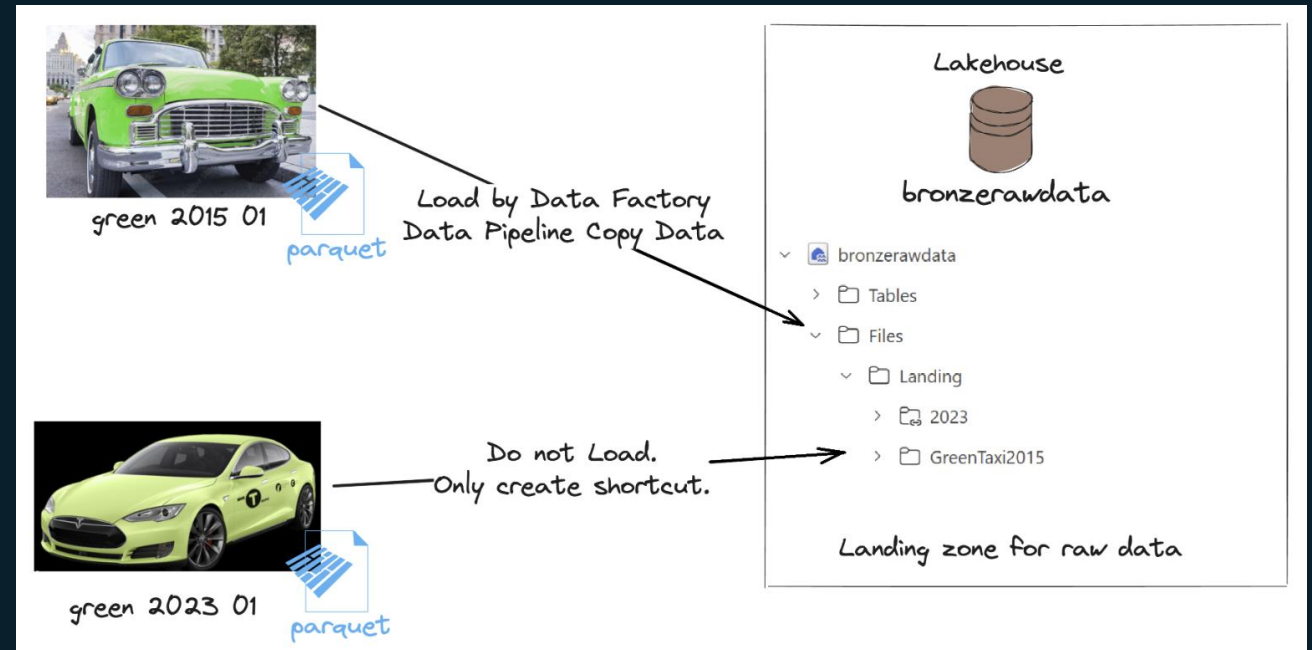
Search tables

bronzerawdata

- Tables
- Files
 - Landing
 - 2023
 - GreenTaxi2015

Files > Landing > GreenTaxi2015 > green_tripdata_2015-01.parquet (preview)

	12%	VendorID	lpep_pickup...	lpep_dropof...	ARC_store_and_f...	12%	RatecodeID	12%	PL
1	2		2015-01-01T00...	2015-01-01T00...	N	1			255
2	2		2015-01-01T00...	2015-01-01T00...	N	1			75
3	2		2015-01-01T00...	2015-01-01T00...	N	1			43
4	2		2015-01-01T00...	2015-01-01T00...	N	1			80
5	2		2015-01-01T00...	2015-01-01T00...	N	1			37
6	2		2015-01-01T00...	2015-01-01T01...	N	1			80
7	1		2015-01-01T00...	2015-01-01T01...	N	1			25



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Ch03: Load bronze tables

Goal

In this challenge, you will load three files into three Delta tables in the [bronze layer](#) lakehouse. You will use three different load strategies:

1. ETL Developer: Data Factory Pipeline
2. Citizen Data Developer: Data Factory Dataflow Gen2
3. Data Engineer: Notebook

Actions

• ETL Developer: Data Factory Pipeline

- Create a new pipeline: Load Green Taxi 2015
- Source:
 - Connection: bronzeraawdata lakehouse
 - Root folder: Files
 - File path: Landing/GreenTaxi2015/green_tripdata_2015-01.parquet
 - File format: Parquet
- Destination:
 - Connection: bronzeraawdata lakehouse
 - Root folder: Tables
 - Table: green201501 (new table)
 - Table action: Overwrite

• Citizen Data Developer: Data Factory Dataflow Gen2

- Create a new Dataflow Gen2: Load Green Taxi Discounts
- Source: bronzeraawdata lakehouse
 - File path: Files > Landing > 2023 > NYC-Taxi-Discounts-Per-Day.csv
- Destination: bronzeraawdata lakehouse
- Table: greenDiscountsPerDay

• Data Engineer: Notebook

- Create a new Notebook: Load Green Taxi 2023 or Import [challenges/ch03/Load Green Taxi 2023 - Ch03.ipynb](#)
 - Remove origin lakehouses from Notebook and Add/attach your bronzeraawdata (as default) lakehouse
- Source: Files/Landing/2023/green_tripdata_2023-01.parquet
- Target: Tables/green202301
 - Format: delta
 - Mode: overwrite

- Check # of rows per delta tables

Success Criteria

- Table green201501 exists with 1 508 493 rows.
- Table green202301 exists with 68 211 rows.
- Table greenDiscountsPerDay exists with 2 rows.

Explorer

Data Items Resources

+ Add data items

Items

bronzeraawdata

Tables

green201501

green202301

greenDiscountsPerDay

```
1 df_green201501 = spark.read.format("delta").load("Tables/green201501")
2 df_green201501 = spark.read.format("delta").load("Tables/green201501")
3 df_green201501 = spark.read.format("delta").load("Tables/green201501")
4
5 print("green201501 row count:", df_green201501.count())
6 print("green201501 row count:", df_green201501.count())
7 print("greenDiscountsPerDay row count:", df_greenDiscountsPerDay.count())
```

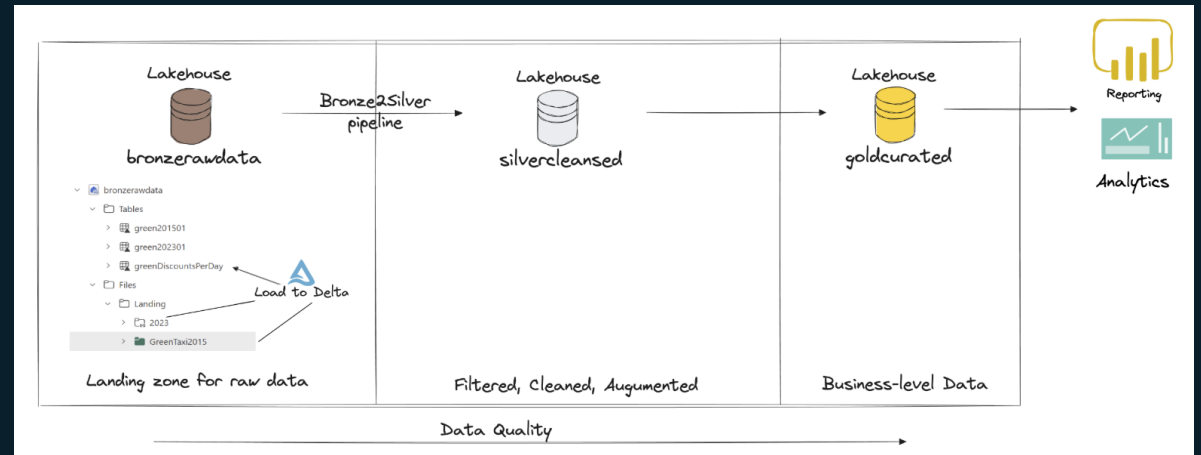
8 sec - Command executed in 8 sec: 30 ms by System Administrator on 5/28/24 PM, 10/07/25

Spark jobs (8 of 8 succeeded) Resources

green201501 row count: 68211

green201501 row count: 1508493

greenDiscountsPerDay row count: 2



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Ch04: Transform data using notebooks and Spark (silver tables)

Goal

In this challenge, you will explore data engineering tasks aimed at transforming raw data into a refined [silver layer](#).

By the end of the challenge, you will have completed the second step, the silver layer, of the [medallion architecture](#):

Actions

- Create the 2nd lakehouse: **silvercleansed**
- Import [challenges/ch04/Transform bronze to silver - Ch04.ipynb](#) to your workspace and try to complete all Goals / Actions mentioned in notebook
 - Remove origin lakehouses from Notebook and Add/attach your **bronzerawdata** (as default) lakehouse and **silvercleansed** lakehouse

Success Criteria

- All success criteria mentioned in the notebook are met
- 6 tables with data in **silvercleansed** lakehouse exist

Explorer

Data Items Resources

+ Add data items

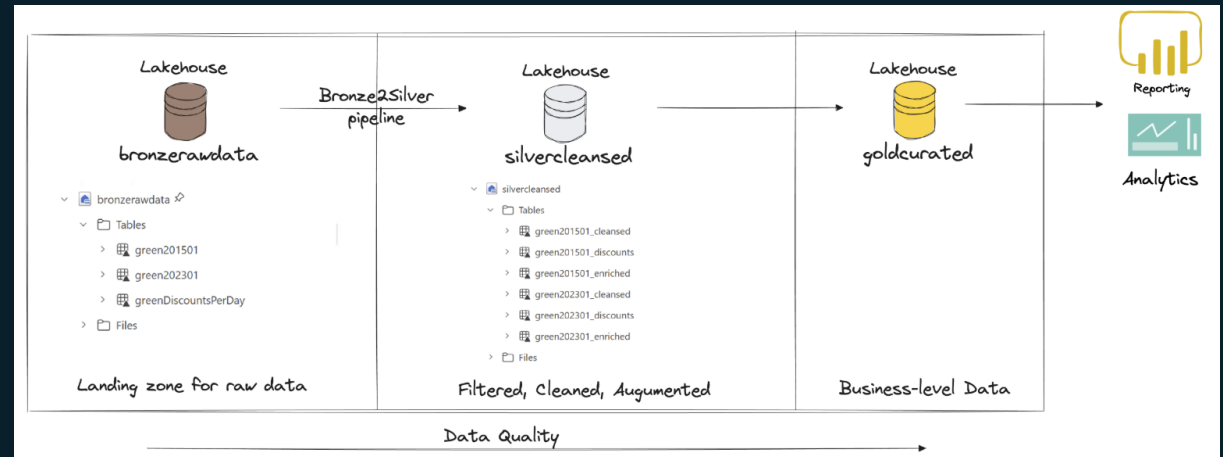
Items

- bronzerawdata
- silvercleansed
 - Tables
 - green201501_cleansed
 - green201501_discounts
 - green201501_enriched
 - green202301_cleansed
 - green202301_discounts
 - green202301_enriched
 - Files

Written 1476971 records

Table view + New chart

	VendorID	lpep_pickup_datetime	PU.LocationID	DOL.LocationID	RateCodeID	passenger_count	trip_distance	trip_duration	pick
1	2	2015-01-01 00:31:10	255	234	1	1	5.88	19.5	2015-01-01
2	2	2015-01-01 00:01:05	75	74	1	1	0.89	2.4	2015-01-01
3	2	2015-01-01 00:09:01	43	186	1	1	5.71	24.4	2015-01-01
4	2	2015-01-01 00:17:34	80	36	1	1	1.89	9.6	2015-01-01
5	2	2015-01-01 00:32:38	37	17	1	1	1.07	7.9	2015-01-01
6	2	2015-01-01 00:45:30	80	97	1	1	2.33	15.5	2015-01-01
7	1	2015-01-01 00:54:04	25	7	1	1	10.8	21.8	2015-01-01
8	1	2015-01-01 00:07:16	223	7	1	1	0.7	3.6	2015-01-01
9	1	2015-01-01 00:14:30	7	146	1	1	1.0	4.4	2015-01-01
10	1	2015-01-01 00:40:52	223	7	1	1	1.1	5.6	2015-01-01
11	1	2015-01-01 00:50:57	7	36	1	2	7.4	19.2	2015-01-01
12	2	2015-01-01 00:57:48	80	225	1	1	4.07	22.5	2015-01-01
13	2	2015-01-01 00:36:14	7	179	1	2	0.76	3.5	2015-01-01
14	2	2015-01-01 00:46:00	7	234	1	3	4.45	28.9	2015-01-01



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Ch05: Build gold tables and a semantic layer for Power BI reporting

Goal

In this challenge, you will aggregate augmented data from a [silver layer](#) into a curated [gold layer](#) for advanced business analysts. Next, you will define semantic layer for end users and create the first Power BI report.

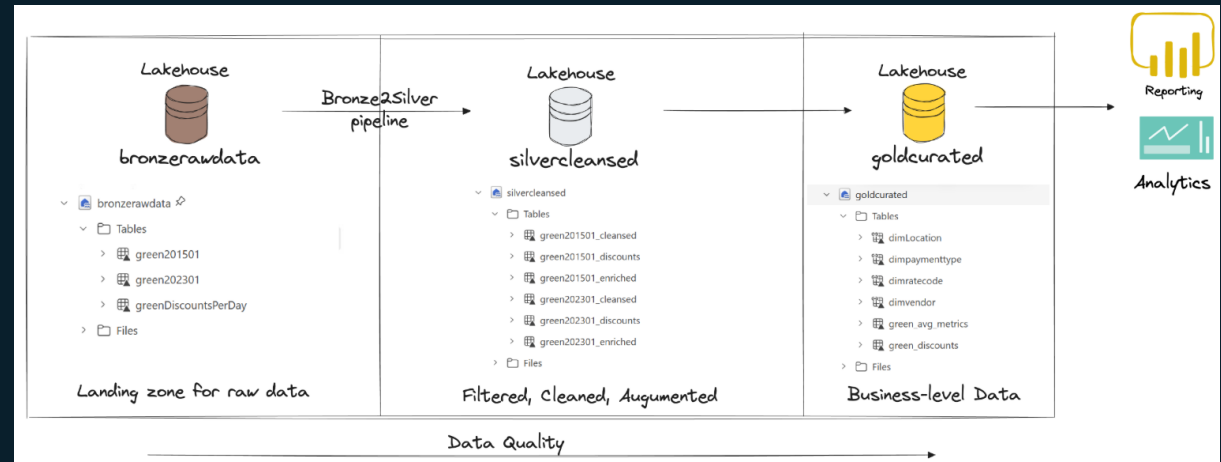
By the end of the challenge, you will have completed the implementation of the [medallion architecture](#) with semantics layer and reports:

Actions

- Create the 3rd lakehouse: **goldcurated**
- Create Shortcut in your Lakehouse into **Tables** area to pre-created delta tables: **dimLocations**, **dimpaymenttype**, **dimratecode**, **dimvendor** in Azure Data Lake Gen2:
 - <https://adlstmicrohacksdfs.core.windows.net/>
 - Authentication kind: **Organizational account**
- Import [challenges/ch05/Transform silver to gold - Ch05.ipynb](#) to your workspace and try to complete all Goals / Actions mentioned in notebook
 - Remove origin lakehouses from Notebook and Add/attach your **silvercleansed** (as default) lakehouse and **goldcurated** lakehouse
- Build Semantics model **GreenTaxi** based on 4 dimensions and 2 facts tables with Avg. function and relations
 - dimLocations.LocationID 1:N green_discount.PULocationID
 - dimpaymenttype.payment_type 1:N green_discount.payment_type
 - dimratecode.RatecodeID 1:N green_discount.RateCodeID
 - dimvendor.VendorID 1:N green_discount.VendorID
- dimLocations.LocationID 1:N green_avg_metrics.PULocationID
- dimvendor.VendorID 1:N greengreen_avg_metrics.VendorID
- Create sample report based on **GreenTaxi** Semantic model with
 - 1 Clustered bar chart, 1 Line chart, 1 Azure maps, 1 Narrative and 1 Slicer
- Create SQL View **vGetAverageFaresDiscounts** based on "Lakehouse SQL analytics endpoint" for business analysts
 - Get the distinct passenger count, total fares, average fares and discount grouped by the passenger count

Success Criteria

- All success criteria mentioned in the notebook are met
- 6 tables with data in **goldcurated** lakehouse exist
- Semantics model **GreenTaxi** with 2 Fact tables and 4 Dimensions exists
- Report based on **GreenTaxi** Semantics model exist
- SQL View **vGetAverageFaresDiscounts** returns 8 records



MicroHack: Microsoft Fabric Lakehouse End-to-End

Ch06: Create a master pipeline for orchestration

Goal

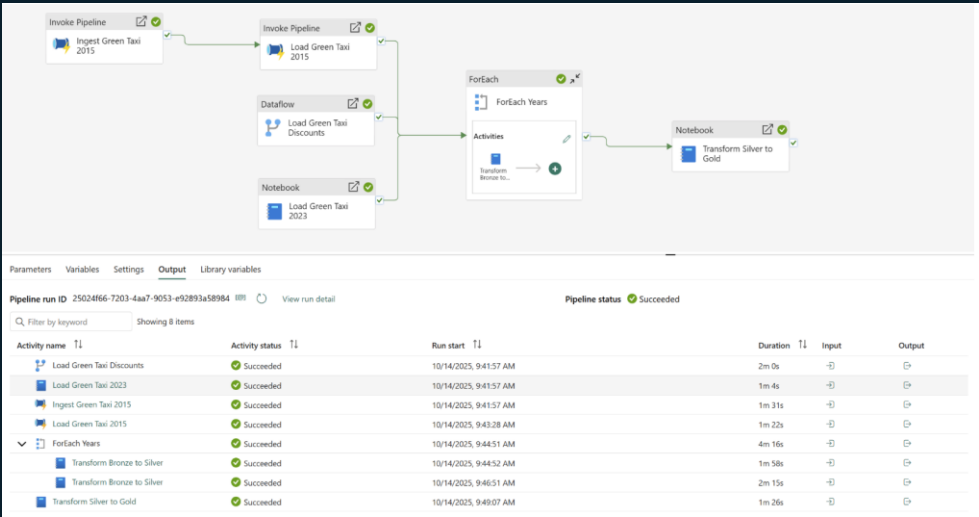
Now, let's focus on orchestration. Considering we have only two tables ingested by two pipelines and processed by three notebooks, imagine a scenario where you need to ingest and process tens or hundreds of tables, files, or data sources and manage dependencies between them. In such cases, the most efficient approach is to build and prioritize a data pipeline for the orchestration workflow.

Actions

- Create master Orchestration Pipeline
 - Add pipeline parameter: Year (Array, Default value = ["2015", "2023"])
 - Build following workflow:
 - Run in parallel:
 - Invoke pipeline: Ingest Green Taxi 2015
 - On Success, run: Invoke Load Green Taxi 2015
 - Run Dataflow: Load Green Taxi Discount
 - Run Notebook: Load Green Taxi 2023
 - On Success, run: ForEach activity: Sequential | Parallel for @pipeline().parameters.Year
 - Run Notebook: Transform Bronze to Silver
 - Add Base parameter p_year binded to ForEach items.
 - On Success: Run Notebook: Transform Silver to Gold
- Run pipeline

Success Criteria

- Pipeline finishes successfully



MicroHack: Microsoft Fabric Lakehouse End-to-End

Ch07: Process unstructured data with an LLM

Goal

In this challenge, you will focus on processing unstructured data from real user reviews of locations in New York.

With Microsoft Fabric, all business professionals—data engineers, data scientists, developers, and analysts—can gain more value from their data through Generative AI.

There are two features in Microsoft Fabric today:

1. Use [AI services](#) with prebuilt and customizable APIs and models, but you need to develop code to process the output into the final schema:
 - Language Models: gpt-5, gpt-4.1, gpt-4.1-mini, and gpt-4o-mini
 - APIs: Text Analytics, and Azure AI Translator
2. Use the new set of [AI functions](#) for data engineering, which is much easier. Fabric users can take advantage of industry-leading large language models (LLMs) to transform and enrich data with just a single line of code.

By the end of this challenge, you will have a new dataset with sentiment, classification, summarization, translation, key topics, and location extracted from real user reviews.

Tip

It's up to you which feature you choose, but this challenge is designed for AI functions.

Actions

There are two options, (1) without Data Preparation tasks and (2) with Data Preparation tasks:

1. without Data Preparation tasks:

- You can create new lakehouse for Sandboxing or use your **goldcurated** lakehouse.
- Create Shortcut in your Lakehouse into **Tables** area to pre-created delta table: **NY_places_customer_reviews_parsed** from Azure Data Lake Gen2 (Container: **reviews**):
 - <https://adlstmfabricmicrohacksw.dfs.core.windows.net/>
 - Authentication kind: **Organizational account**
- Import [challenges/ch07/User reviews processing with LLM - AI Functions - Ch07.ipynb](#) to your workspace and try to complete all Goals / Actions mentioned in that notebook
 - Remove origin lakehouses from Notebook and Add/attach your **goldcurated** (as default) lakehouse

2. with Data Preparation tasks:

- You can create new lakehouse for Sandboxing or use your **goldcurated** lakehouse.
- Upload file [NY_places_customer_reviews.csv](#) to your lakehouse (Files area)
- Import [challenges/ch07/User reviews processing with LLM - AI Functions with DE - Ch07.ipynb](#) to your workspace and try to complete all Goals / Actions mentioned in that notebook
 - Remove origin lakehouses from Notebook and Add/attach your **goldcurated** (as default) lakehouse

Success Criteria

- All success criteria mentioned in the notebook are met
- Table **goldcurated.reviewLocations** exists with 265 rows and AI columns

```
1 df = spark.sql("SELECT * FROM goldcurated.reviewLocations LIMIT 3000")
2 display(df)
```

✓ - Command executed in 29 sec 67 ms by System Administrator on 2/12/25 PM 10/16/25

Table

New chart

Table view

Id	LocationID	Place	Review	User	Sentiment	Classify	Summarize	TranslationsReview	TranslationsSummarize	KeyTopics	Locations
1	100	New York	Users Island	Anonymous	neutral	historical building	Users Island is NY	Users Island je historická vísť	Users Island je historická vísť	Users Island je historická vísť	Users Island, NYC, Queens

```
1 # TRANSLATE input text to Czech language
2
3 dfp["AItranslationsReview"] = dfp["Review"].ai.translate("Czech")
4 dfp["AItranslationsSummarize"] = dfp["AIsummarize"].ai.translate("Czech")
5 display(dfp)
6
```

✓ - Command executed in 18 sec 692 ms by System Administrator on 3:20:03 PM, 10/24/25

100%|██████████| 265/265 [00:07<00:00, 36.36it/s]

100%|██████████| 265/265 [00:06<00:00, 39.93it/s]

Table

New chart

Table view

	Id	LocationID	Place	Review	User	Sentiment	Classify	Summarize	TranslationsReview	TranslationsSummarize
1	1		New York: ...	Terrible if yo...	Bocodude	negative	historic building	User describes a ...	Hrozné, pokud musíte m...	Uživatel popisuje velké ameri...
2	2		New York: ...	Even in centr...	Hoben01	positive	accommodation	In central NYC, yo...	I v centru New Yorku si st...	V centru New Yorku můžete ...
3	3		New York: ...	Pelham Gard...	Joe Cosenti...	positive	accommodation	Pelham Gardens is ...	Pelham Gardens je obytn...	Pelham Gardens je centrálně ...
4	4		New York: ...	Alphabet Cit...	Lula N	positive	bars	Alphabet City is a ...	Alphabet City působí jako...	Alphabet City je skutečná čtv...
5	5		New York: ...	Arden Heigh...	Anonymous	positive	accommodation	Arden Heights is ...	Arden Heights je domove...	Arden Heights je předměstsk...
6	6		New York: ...	Beautiful vie...	Michael S	mixed	historic building	Beautiful views an...	Krásné výhledy a zajímav...	Krásné výhledy a zajímavá hi...
7	7		New York: ...	Astoria is a v...	Current Re...	positive	restaurants	Astoria is a safe, ...	Astoria je velmi pěkná čtv...	Astoria je bezpečná, dobře o...

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Ch08: Data Agent: Talk with your data

Goal

In this challenge, you will focus on the Data Agent, which allows you to build your own conversational Q&A systems using generative AI. The Fabric Data Agent uses LLMs to help users interact with their data naturally. It leverages Azure OpenAI Assistant APIs and behaves like an agent: processing user questions, determining the most relevant data source (Lakehouse, Warehouse, Power BI dataset, KQL databases), and invoking the appropriate tool to generate, validate, and execute queries. This way, even people without technical expertise in AI or a deep understanding of the data structure can receive precise and context-rich answers.

During this challenge, you will define a new Data Agent and add specific instructions, examples, and guidance to fine-tune the Fabric Data Agent. This ensures that responses align with your needs and goals, allowing everyone to engage with data more effectively.

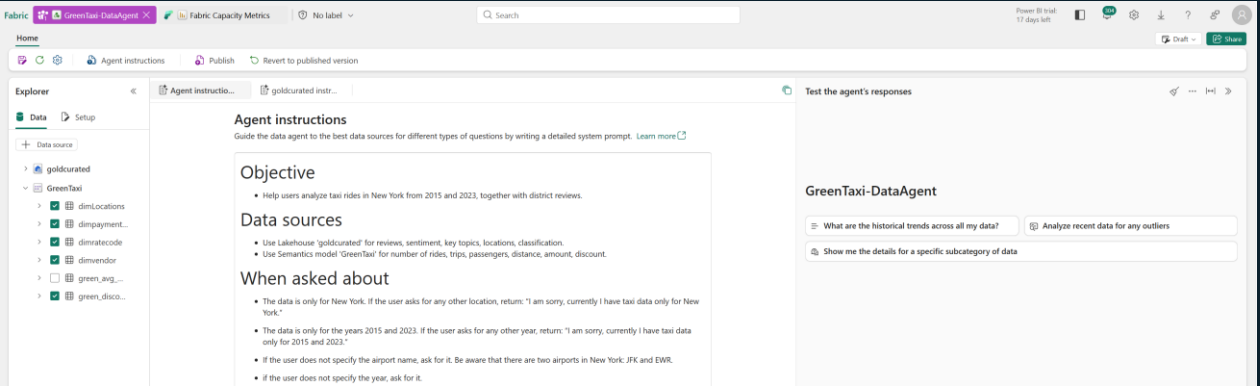
By the end of this challenge, the Data Agent will be ready to be consumed via the Python SDK, Azure AI Foundry, Microft Copilot Studio, and Copilot in Power BI.

Actions

- Create new Data Agent
- Add Data sources:
 - Semantic model: GreenTaxi
 - Tables:
 - dimLocations
 - dimpaymenttype
 - dimratecode
 - dimvendor
 - green_discount
 - Lakehouse: goldcurated
 - Tables:
 - dimLocations
 - reviewLocations
- Test questions:
 - What were the most common pickup locations for green taxi rides in 2023?
 - On which weekday do people travel from the airport the most?"
 - How many rides were in Boston?
 - How many rides were there in 2025?
 - Show me the top 3 districts in New York with the best accommodation.
 - Show me the districts where the most passengers moved from.
 - Show me the districts where most passengers moved from, and what reviews, sentiments, and key topics you know about these districts.
- Tune Data Agent via Agent instructions (System Prompt) and Data source instructions for above questions
- Publish Data Agent
- Consume Data Agent via Copilot (Home) in Power BI or PythonSDK

Success Criteria

- Data Agent exists and is able to answer questions about data in Semantics model and Lakehouse.
- Data Agent is available via Python SDK or Copilot (Home) in Power BI
- Data Agent can answer questions:
 - Question: What were the most common pickup locations for green taxi rides in 2023?
 - Answer: East Harlem North; East Harlem South; Forest Hills; Central Harlem; ...



Let's Start Hacking



MicroHack-Fabric GitHub Repo:

<http://aka.ms/MicroHackFabric>
(<https://github.com/CZSK-MicroHacks/MicroHack-Fabric>)

